DNS

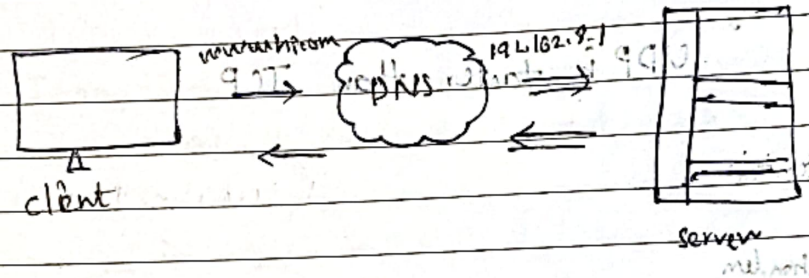
↳ Domain name system

** DNS is a system that converts domain names into Ip address..

* Human cannot remember all application Ip address where the Internet & machine network communicates with the Ip address

→ where each system has their own unique Ip address.

Human just known the google.com, Youtube.com, amazon.com } ⇒ Just the domain Names
⇓
but Internet communicates with DNS



* DNS is a distributed system that maps domain names to Ip address so computer can locate on internet..

without DNS → '142.250.193.14' ⇒ remembering this Ip address..

This Number cannot remembered so the DNS conversion can be done

So this core idea.

when ever the user click (or) enter an www.google.com
It will first go to the DNS server &
then it will go to the Server.....

DNS Resolution:-

* where ever we type → www.google.com
↓
The system will find the IP address through several steps.

① Browser cache:-

The browser chache is an Browser cache memory which the browser check if the IP address is already stored.

If found → connect start immediately.

② Operating system cache:-

If the browser doesn't have it, the OS cache is then checked.

Ping google.com

③ DNS resolver:- If the OS also doesn't know the request goes to "DNS Resolver"

Example

google DNS → 8.8.8.8

The resolver find the IP address by querying other DNS server

DNS resolver
↓
Root DNS server → Ask

④ The root DNS server

The root DNS server doesnot know the exact IP but it will tell where to find the IP...

⑤

Eg Ask the .com TLD server
↓
Top level domains

TLD Domain
↓
.com
.org
.net
.edu

→ The TLD server responds:
Ask the authoritative server for google.com

(6) Authoritative DNS Server:

The resolver asks the authoritative server
This server contains the actual record..

google.com → 142.250.193.14..

(7) Response returned... → The resolver sends the IP address back to browser

Actual procedure

Type google.com
↓
Browser checks DNS cache
↓
OS cache checked
↓
DNS resolver
↓
Resolver queries DNS root server
↓
Root server's direct to TLD server
↓
TLD server points to authoritative server
↓
Browser connects to server
↓
HTTP request is sent...

DNS caching :-

To make the internet faster DNS results are cached..

google → 142.250.193.14

This mapping is stored temporarily... → handled by TTL

After TTL expires, DNS must resolve again...

This whole cycle will take an extra link..

DNS resolver: From the Initial query to the Root DNS server.

Types of DNS server:-

① Root DNS server:- Top level of the DNS hierarchy
Direct queries to the correct TLD server

② Top level Domain server → TLD:-
Handles domain endings
.com, .org, .edu, .in

③ Authoritative server:
stores the actual domain → IP mapping.

amazon.com → 205.251.242.103

"By this actual implementation done"

Why does Authoritative server have ...?
when ever the domain is bought they will say the register domain names into it..